OSCAR ROBERT OKIYA

Cloud Infrastructure Engineer | Solution Design & Delivery | Multi-Cloud (AWS/GCP) & Kubernetes

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PROFESSIONAL SUMMARY

Cloud infrastructure engineer with 5+ years scoping, designing, and delivering resilient platform solutions in high-availability fintech environments. Have acted as one of the lead architects on two end-to-end infrastructure modernisation programmes — analysing requirements, assessing risk, and driving solutions from design through implementation, verification, and rollout — for systems supporting 200,000+ end users. Strong hands-on Kubernetes (EKS/GKE), Linux, and Infrastructure-as-Code background, with a track record of translating technical decisions into measurable business outcomes (30% cost reduction, 50% faster delivery, 90% improvement in payment accuracy). Skilled in stakeholder management and customer-facing delivery — developing technical proposals and design documentation, presenting migration approach and trade-offs directly to client stakeholders, and communicating system design methodology to 500+ engineers at Africa DevOps Summit 2025. Holds three AWS Professional/Specialty certifications and is actively building OpenStack infrastructure competence to extend a strong public-cloud foundation.

CORE TECHNICAL EXPERTISE

Container Orchestration: Kubernetes (EKS, GKE) — deployments, scaling, service networking; Docker; ECS, Cloud Run

Cloud Platforms: AWS (EC2, Lambda, S3, RDS, VPC, ECS, Cognito, ElastiCache, CloudFormation, CodePipeline); GCP (Compute Engine, Cloud Run, BigQuery, Cloud SQL, Pub/Sub, GKE, Cloud Build)

Linux & Networking: Linux server administration; VPC/VPN design, CloudFront, load balancing, network troubleshooting; storage design (RDS, S3, EBS, Cloud SQL)

IaC & Automation: Terraform, Terraform Cloud, AWS CloudFormation, AWS SAM, Bash, Python, Git

CI/CD & Delivery: GitHub Actions, GitLab CI, AWS CodePipeline, Cloud Build, trunk-based development, Apigee

Observability & Capacity Planning: Prometheus, Grafana, OpenSearch, CloudWatch, Cloud Monitoring, SLI/SLO/SLA definition, DORA metrics, demand forecasting

Security & Secrets: IAM, KMS, CloudTrail, HashiCorp Vault, Consul, IAP, Security Command Centre

Resilience Engineering: Chaos Engineering (Gremlin), disaster recovery design, high-availability architecture

PROFESSIONAL EXPERIENCE

Senior DevOps Engineer (Acting Infrastructure Solution Architect) — 4G Capital, Nairobi *Mar 2022 – Present*

- Scoped, designed, and co-led delivery of two distinct infrastructure modernisation solutions end-to-end: analysed business requirements and performed risk assessment to migrate AWS ECS workloads to a serverless model (AWS SAM/CloudFormation), then architected a full cross-cloud migration to GCP (Cloud Run) using Terraform — reducing cloud costs by 30% and eliminating all manual (ClickOps) infrastructure changes
- Authored technical proposals and design documentation for infrastructure initiatives — presenting architecture options, risk trade-offs, and cost impact to engineering leadership, and translating the same decisions for internal customers across development, product, and support teams to secure buy-in before implementation
- Defined and drove a 50% acceleration in software delivery by designing unified CI/CD solution architecture across AWS and GCP, leading stakeholder alignment on a shift from GitFlow to trunk-based development, and standardising artifact promotion across all environments
- Designed and implemented a payment validation solution that reduced reconciliation errors by 90%, applying structured root-cause analysis to a system where accuracy directly affected financial outcomes
- Architected a high-concurrency messaging platform supporting expansion to 200,000+ customers within a 5-minute delivery window, incorporating capacity planning, load testing, and cost-per-broadcast reporting for management visibility
- Built full-platform observability practice from the ground up — defining SLIs, SLOs and SLAs, and delivering DORA metrics and OpenSearch dashboards that directly informed capacity and demand-planning decisions for engineering leadership
- Governed infrastructure change management by introducing Terraform Cloud, enforcing all changes through reviewed, versioned, and audited pipelines — a process-improvement initiative that removed developer-side state conflicts

- Acted as the technical owner accountable for production reliability across the platform, with development, product, and support teams as direct internal customers depending on uptime, performance, and incident response; led knowledge transfer and mentorship for 4 junior engineers, improving onboarding speed by 40%

DevOps Engineer — Ajua, Nairobi *Jun 2021 – Jun 2022*

Continued part-time from Mar 2022 to complete ongoing infrastructure projects

- Modernised the infrastructure platform by migrating all services from EC2 to AWS ECS, introducing containerisation to standardise deployments and automate rollbacks
- Configured HashiCorp Vault and Consul for secrets management and service discovery; adopted Terraform for all cloud provisioning, improving reproducibility and eliminating manual configuration errors
- Consolidated multi-region infrastructure (N. Virginia and Oregon into Ireland), rightsizing compute and introducing CloudFront — cutting service latency by 75% and reducing infrastructure spend by 40%, directly supporting clients across multiple African markets
- Migrated PostgreSQL from self-managed EC2 to AWS RDS with replication and consolidated VPN connectivity for distributed clients into a single managed, scalable solution — improving resilience and reducing operational risk
- Designed CI/CD pipelines for Java and Node.js applications across EC2 and ECS deployment targets, coordinating environment-specific release processes with development stakeholders
- Authored infrastructure and runbook documentation for development teams — the platform's direct internal customers — while holding accountability for production availability that the wider business depended on for continuous operation

AWS Cloud Consultant — Recon AI, Helsinki (Remote) *Jan 2021 – Mar 2021*

- Engaged directly with an international client to migrate infrastructure from a contractor-managed GCP environment to AWS — gathering requirements, architecting a scalable, resilient solution (CloudFormation, EC2, ALB, Auto Scaling Groups), and authoring migration design documentation that was presented directly to client stakeholders alongside the proposed approach and trade-offs
- Integrated GitLab with AWS to automate CI/CD and implemented a data backup and disaster recovery strategy using S3 and EBS snapshots

SPEAKING & CUSTOMER-FACING ENGAGEMENTS

Speaker — System Design for DevOps Engineers — Africa DevOps Summit 2.0 *2025*

- Authored the session documentation and slide materials and delivered a hands-on technical session to 500+ attendees covering requirements definition, microservices architecture, API and database design, and full cloud deployment for a resilient e-commerce platform at scale (github.com/oscarrobert-star/ecommerce-project)

Guest Speaker — CI/CD Fundamentals — Catholic University of Eastern Africa *2025*

- Presented modern deployment practices, automation, and DevOps principles to computer science students, translating complex technical concepts for a non-specialist audience

CERTIFICATIONS

- AWS Certified Solutions Architect – Professional
- AWS Certified DevOps Engineer – Professional
- AWS Certified Machine Learning – Specialty

EDUCATION

B.Sc. Statistics — Jomo Kenyatta University of Agriculture and Technology, Nairobi (Sep 2015 – Nov 2019)